Foundation of Mathematics

4- Summary for Integration

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1. Rules of Integration

$\int \mathbf{f(x)\,dx}$	$\mathbf{F(x) + c}$
$\int c * f(x)\,dx$	$c * \int f(x)\,dx$ Valid for constant c
$\int (u)^n * u'dx$	$\frac{u^{n+1}}{n+1} + c$; $n \neq -1$
$\int \frac{u'}{u}\,dx$	$\ln\lvert u \rvert + c$
$\int e^u * u'\,dx$	$e^u + c$
$\int a^u * u'\,dx$	$a^u * \frac{1}{\ln a} + c$

$$\mathbf{u'\,dx = du}$$

$$e.g.\ :\ \int (\mathbf{u})^{\mathbf{n}} * \mathbf{u'dx} = \frac{\mathbf{u^{n+1}}}{\mathbf{n+1}} + \mathbf{c\ ;\ n \neq -1}$$

Can be written as:

$$\int (\mathbf{u})^{\mathbf{n}} * \mathbf{du} = \frac{\mathbf{u^{n+1}}}{\mathbf{n+1}} + \mathbf{c\ ;\ n \neq -1}$$

Remarks:

- $\int (f \pm g)\,dx = \int f\,dx \pm \int g\,dx$
- $\int (f * g)\,dx \neq \int f\,dx * \int g\,dx$
- $\int \left(\frac{f}{g}\right) dx \neq \frac{\int f\,dx}{\int g\,dx}$
- $\frac{d}{dx} \int f(x)\,dx = f(x)$
- $\int \frac{df}{dx}\,dx = f(x) + c$

Trignometric Functions

$\int \sin(u) * u'\,dx$	$-\cos(u) + c$
$\int \cos(u) * u'\,dx$	$\sin(u) + c$
$\int \sec^2(u) * u'\,dx$	$\tan(u) + c$
$\int \csc^2(u) * u'\,dx$	$-\cot(u) + c$
$\int \sec(u) * \tan(u) * u'\,dx$	$\sec(u) + c$
$\int \csc(u) * \cot(u) * u'\,dx$	$-\csc(u) + c$

$$\int \sec(x)\,dx = \int \sec(x) * \frac{\sec(x) + \tan(x)}{\sec(x) + \tan(x)} dx$$

$$= \int \frac{\sec^2(x) + \tan(x) * \sec(x)}{\sec(x) + \tan(x)}\,dx$$

$$= \ln[\sec(x) + \tan(x)] + c$$

$$\int \csc(x)\,dx = \int \csc(x) * \frac{\csc(x) - \cot(x)}{\csc(x) - \cot(x)} dx$$

$$= \int \frac{\csc^2(x) - \csc(x) * \cot(x)}{\csc(x) - \cot(x)}\,dx$$

$$= \ln[\csc(x) - \cot(x)] + c$$

$$\int tan^2(x)dx = \int [sec^2(x) - 1]dx$$

$$= \tan(x) - x + c$$

Note:

For $\int \sin^2(x)dx$ **OR** $\int \cos^2(x)dx$

Use : $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$

Or

$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$

$$\int \cos(kx) = \frac{1}{k} * \sin(kx) + c$$

$$\int \sin(kx) = \frac{-1}{k} * \cos(kx) + c$$

$$\int e^{kx} = \frac{1}{k} * e^{kx} + c$$

Hyperbolic Functions

$\int \sinh(u) * u' \; dx$	$\cosh(u) + c$
$\int \cosh(u) * u' \; dx$	$\sinh(u) + c$
$\int \operatorname{sech}^2(u) * u' \; dx$	$\tanh(u) + c$
$\int \operatorname{csch}^2(u) * u' \; dx$	$-\coth(u) + c$
$\int \operatorname{sech}(u) * \tan(u) * u' \; dx$	$-\operatorname{sech}(u) + c$
$\int \operatorname{csch}(u) * \cot(u) * u' \; dx$	$-\operatorname{csch}(u) + c$

$$\int \operatorname{sech}(x)\,dx = \int \frac{2}{e^x + e^{-x}}dx \left[\textbf{multiply by } \frac{e^x}{e^x}\right]$$

$$= \int \frac{2e^x}{e^{2x}+1}dx = 2\int \frac{e^x}{(e^x)^2+1}\,dx = 2\tan^{-1}(e^x) + C$$

$$\int \operatorname{csch}(x)\,dx = \int \frac{2}{e^x - e^{-x}}dx \left[\textbf{multiply by } \frac{e^x}{e^x}\right]$$

$$= \int \frac{2e^x}{e^{2x}-1}dx = -2\int \frac{e^x}{1-(e^x)^2}\,dx = -2\tanh^{-1}(e^x) +$$

Inverse Trignometric Functions

$\int \frac{u' \; dx}{\sqrt{a^2 - u^2}}$	$\sin^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{\sqrt{a^2 - u^2}}$	$-\cos^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{a^2 + u^2}$	$\frac{1}{a} * \tan^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{a^2 + u^2}$	$-\frac{1}{a} * \cot^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{u * \sqrt{u^2 - a^2}}$	$\frac{1}{a} * \sec^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{u * \sqrt{u^2 - a^2}}$	$-\frac{1}{a} * \csc^{-1}(\frac{u}{a}) + C$

Inverse Hyperbolic Functions

$\int \frac{u' \; dx}{\sqrt{a^2 + u^2}}$	$\sinh^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{\sqrt{u^2 - a^2}}$	$\cosh^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{a^2 - u^2}$	$\frac{1}{a} * \tanh^{-1}(\frac{u}{a}) + C$; $u^2 < a^2$
$\int \frac{u' \; dx}{a^2 - u^2}$	$\frac{1}{a} * \coth^{-1}(\frac{u}{a}) + C$; $u^2 > a^2$
$\int \frac{u' \; dx}{u * \sqrt{a^2 - u^2}}$	$-\frac{1}{a} * \operatorname{sech}^{-1}(\frac{u}{a}) + C$
$\int \frac{u' \; dx}{u * \sqrt{a^2 + u^2}}$	$-\frac{1}{a} * \operatorname{csch}^{-1}\left\lvert\frac{u}{a}\right\rvert + C$

Substitution Rule

$$\int f(g(x)) * g'(x)\,dx = \int f(u)\,du$$

where, $g(x) = u$ then, $g'(x)\,dx = du$

2. Determination of the constant of Integration

As we know that

$$\boxed{\int f(x)\,dx = F(x) + c}$$

where, $F(x) + c$ is called family of curves as it depends on the value of the arbitrary constant 'C'.
For real life application, a particular curve will satisfy the conditions of the application. To find that curve, an initial condition must be given to be able to calculate the value of the constant.

3. Definite Integral

Fundamental Theorem Of Calculus Part (I)	$\int_a^b f(x)\,dx = [F(x)]_a^b = F(b) - F(a)$
Order of Integration	$\int_a^b f(x)\,dx = -\int_b^a f(x)dx$
Zero Width Interval	$\int_a^a f(x)\,dx = 0$
Constant Multiple	$\int_a^b K * f(x)\,dx = K * \int_a^b f(x)dx$
Sum and Difference	$\int_a^b [\,f(x) \pm g(x)\,]dx = \int_a^b f(x)dx \pm \int_a^b g(x)dx$
Additivity	$\int_a^b f(x)\,dx + \int_b^c f(x)dx = \int_a^c f(x)\,dx$

Fundamental Theorem Of Calculus Part (II)	**Version (1):** $F'(x) = \frac{d}{dx}\int_a^x f(t)dt = f(x).$ --------------------------- **Version (2):** $\frac{d}{dx}\int_a^{u(x)} f(t)dt = f(u(x)) * u'(x)$ --------------------------- **Version (3):** $\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)dt = f(v) * v'(x) - f(u) * u'(x)$

- If f is even, then $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx.$
- If f is odd, then $\int_{-a}^{a} f(x)dx = 0.$

4. Applications of Definite Integration

4.1. Area Under of a Curve $f(x)$

To find an area bounded by a graph of a function f(x) , where f(x) is non negative and integrable over [a,b]:

- ✓ Check whether f(x) has any x-intercept inside [a,b].
- ✓ If f(x) has no x-intercepts, then :

$$\boxed{\text{Total area} = \left|\int_a^b f(x)dx\right| = |F(b) - F(a)|}$$

- ✓ Otherwise, let $c_1, c_2, \dots, c_n$ be the x-intercepts inside [a,b]. Then,

$$\boxed{\text{Total area} = \left|\int_a^{c_1} f(x)dx\right| + \left|\int_{c_1}^{c_2} f(x)dx\right| + \dots + \left|\int_{c_2}^{b} f(x)dx\right|}$$

4.2. Area between Two Curves $f(x), g(x)$

To find the area bounded between 2 curves, follow the following procedure:

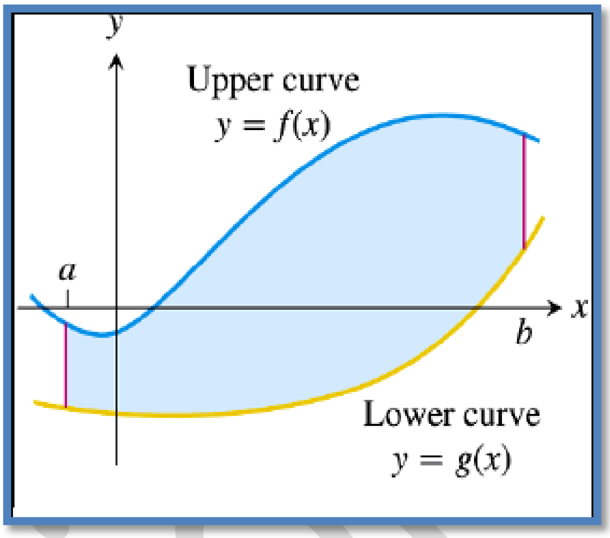
- ✓ Sketch the graph of the given curves $\mathbf{f(x), g(x)}$ and determine the required region.
- ✓ Find the points of intersection between the 2 curves by letting $\mathbf{f(x) = g(x)}$
- ✓ Determine whether you are going to use a **Horizontal** or a **Vertical** strip.
  - For a **Vertical** Strip:

    $$\text{Area} = \int_a^b [Y_{up} - Y_{down}]\, dx$$

  - For a **Horizontal** Strip:

    $$\text{Area} = \int_c^d [X_{right} - X_{left}]\, dy$$

The following figure illustrates the **vertical** strip.



The following figure illustrates the **horizontal** strip.